

MATH 465 - Homework 4 Solutions

Due September 29, 2026

Acknowledgment. I consulted the homework prompt and no other books, notes, or students. This write-up was prepared with generative AI assistance.

Notation. The signless Stirling number $c(n,k)$ counts permutations of $\{1,\dots,n\}$ having exactly k cycles. The Stirling number of the second kind $S(n,k)$ counts partitions of an n -element set into exactly k nonempty unlabeled blocks.

1. A recurrence for $c(n,k)$

Consider a permutation of $\{1,\dots,n\}$, and distinguish the element n . If n is a one-element cycle, removing it leaves a permutation of $\{1,\dots,n-1\}$ with $k-1$ cycles, giving $c(n-1,k-1)$ possibilities.

Otherwise, start with a permutation of $\{1,\dots,n-1\}$ with k cycles and insert n immediately after one of the existing $n-1$ elements in its cycle notation. There are $n-1$ choices, and the number of cycles is unchanged, giving $(n-1)c(n-1,k)$ possibilities.

The two cases are disjoint and exhaustive. Therefore

$$c(n,k) = c(n-1,k-1) + (n-1)c(n-1,k).$$

2(a). $c(n,n-1) = \text{binom}(n,2)$

A permutation of n elements with $n-1$ cycles has cycle deficit one. Thus exactly one pair is combined into a 2-cycle, while every other element is fixed. Choosing the pair gives $\text{binom}(n,2)$ possibilities, and the 2-cycle is then determined.

$$c(n,n-1) = \text{binom}(n,2).$$

2(b). $c(n,n-2) = 2 \text{binom}(n,3) + 3 \text{binom}(n,4)$

The cycle deficit is two. There are two possible structures: one 3-cycle and all other elements fixed, or two disjoint 2-cycles and all other elements fixed.

For the first structure, choose the three elements and then choose one of the $(3-1)! = 2$ cycles on them. This contributes $2 \text{binom}(n,3)$. For the second, choose four elements and split them into two pairs. There are 3 pairings of four specified elements, contributing $3 \text{binom}(n,4)$.

$$c(n,n-2) = 2 \text{binom}(n,3) + 3 \text{binom}(n,4).$$

2(c). $S(n,n-2) = \text{binom}(n,3) + 3 \text{binom}(n,4)$

For a partition into $n-2$ blocks, the block-size deficit from all singletons is two. Hence either one block has size three, or two blocks have size two.

A size-three block can be chosen in $\text{binom}(n,3)$ ways. For two size-two blocks, choose four elements and pair them; there are 3 pairings. Therefore

$$S(n,n-2) = \text{binom}(n,3) + 3 \text{binom}(n,4).$$

3. Tourists at circular tables with drinks

Let T_n count the arrangements, and set $T_0=1$ for the empty arrangement. Given an arrangement of $n-1$ tourists, tourist n either forms a new one-person table, with r choices for its drink, or joins an existing table. In the latter case, choose the tourist immediately to the right of n ; there are $n-1$ choices, and the table keeps its drink.

Conversely, deleting n recovers the unique preceding arrangement, so this construction is bijective. Hence

$$T_n = (r+n-1)T_{n-1}, \quad T_0=1.$$

Iterating the recurrence gives

$$T_n = r(r+1)\dots(r+n-1).$$

4. A hockey-stick identity

Count the $(k+1)$ -element subsets of $\{1,2,\dots,n+1\}$. Classify a subset by its largest element. If the largest element is $j+1$, where $k \leq j \leq n$, then the other k elements can be chosen from $\{1,\dots,j\}$ in $\text{binom}(j,k)$ ways.

Thus the subsets are counted by the sum below. Directly choosing any $(k+1)$ -element subset gives the second count, so

$$\sum_{j=k}^n \text{binom}(j,k) = \text{binom}(n+1,k+1).$$

This is a direct double count and does not use Pascal's recurrence or algebraic manipulation of binomial coefficients.

5. The sum of cubes

Use the identity $j^3 = 6 \text{binom}(j,3) + 6 \text{binom}(j,2) + \text{binom}(j,1)$, together with the identity from Problem 4.

$$\begin{aligned} \sum_{j=0}^n j^3 &= 6 \text{binom}(n+1,4) + 6 \text{binom}(n+1,3) + \text{binom}(n+1,2) \\ &= n^2(n+1)^2/4 \\ &= [n(n+1)/2]^2. \end{aligned}$$

6(a). A third-order recurrence

The characteristic equation is

$$\lambda^3 - 2\lambda^2 - \lambda + 2 = (\lambda-2)(\lambda-1)(\lambda+1).$$

Therefore $a_n = A2^n + B + C(-1)^n$. The initial values give $A+B+C=4$, $2A+B-C=8$, and $4A+B+C=10$. Solving gives $A=2$, $B=3$, $C=-1$.

$$a_n = 2^{n+1} + 3 - (-1)^n.$$

6(b). A repeated characteristic root

The characteristic equation is

$$\lambda^3 - 9\lambda^2 + 27\lambda - 27 = (\lambda-3)^3.$$

Thus $a_n = (A+Bn+Cn^2)3^n$. The initial values give $A=1$, $B+C=0$, and $2B+4C=2$. Hence $B=-1$ and $C=1$.

$$a_n = (n^2 - n + 1)3^n.$$

7. A recurrence for $a_n = (2n-1)3^n + 4(-1)^n$

The term $(2n-1)3^n$ is a linear combination of 3^n and $n3^n$, so the characteristic root 3 has multiplicity two. The term $(-1)^n$ contributes the root -1.

$$(\lambda - 3)^2(\lambda + 1) = \lambda^3 - 5\lambda^2 + 3\lambda + 9.$$

Therefore one suitable recurrence is

$$a_n = 5a_{n-1} - 3a_{n-2} - 9a_{n-3}, \quad n \geq 3.$$

8. A Fibonacci sum

Tile a board of length n with monomers of length one and dominoes of length two. If T_n is the number of tilings, looking at the first tile gives $T_n = T_{n-1} + T_{n-2}$, with $T_0 = T_1 = 1$. Thus $T_n = f_{n+1}$.

If a tiling contains exactly k dominoes, then it contains $n-2k$ monomers and $n-k$ tiles in total. Choosing which k tile positions are dominoes gives $\binom{n-k}{k}$ tilings. Therefore

$$f_{n+1} = \sum_{k=0}^{\lfloor n/2 \rfloor} \binom{n-k}{k}.$$

9. The Fibonacci addition identity

Consider tilings of a board of length $m+n$ and the boundary between positions m and $m+1$. If no domino crosses the boundary, the two segments can be tiled independently, giving $f_{m+1}f_{n+1}$ tilings.

If a domino crosses the boundary, remove it. The remaining segments have lengths $m-1$ and $n-1$, giving $f_m f_n$ tilings. These classes are disjoint and exhaustive.

$$f_{m+n+1} = f_{m+1}f_{n+1} + f_m f_n.$$

10(a). The Fibonacci generating function

Let $F(x) = \sum_{n \geq 0} f_n x^n$. Using the recurrence and initial values,

$$F(x) = x + xF(x) + x^2F(x).$$

$$F(x) = x/(1-x-x^2).$$

10(b). Compositions into odd parts

The generating function for one positive odd part is $x+x^3+x^5+\dots = x/(1-x^2)$. A nonempty composition is a nonempty sequence of such parts, so

$$A(x) = [x/(1-x^2)]/[1-x/(1-x^2)] = x/(1-x-x^2).$$

This equals $F(x)$, so coefficient comparison gives

$$a_n = f_n, \quad n \geq 1.$$

10(c). A bijection

Write each odd part as $2q+1$ and replace it by q copies of 2 followed by one 1. Concatenate the replacements and delete the final 1. The total decreases by exactly one, producing a composition of $n-1$ into parts 1 or 2.

$$(5, 3, 1) \rightarrow (2, 2, 1, 2, 1, 1) \rightarrow (2, 2, 1, 2, 1).$$

For the inverse, append a final 1 to a composition of $n-1$ into 1s and 2s, then split immediately after each 1. Each block has the form $(2, \dots, 2, 1)$, so it corresponds uniquely to an odd part. The two operations are inverse. For $n=1$, the empty composition of 0 is used.

Compositions of $n-1$ into 1s and 2s are tilings of a board of length $n-1$ by monomers and dominoes, so they are counted by f_n . The bijection therefore proves again that $a_n = f_n$.